The Influence of Adult Children's Education on the Health Behavior of Parents

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Abstract: China faces challenges in promoting "healthy aging", particularly with the world's largest elderly population. The health issues of older demographics require more attention. This article analyzes the health behaviors of middle-aged and elderly groups using data from the China Family Panel Studies (CFPS) from 2010 to 2018. The study uses instrumental variables to investigate the impact of adult children's educational attainment on the health behaviors of their older parents. Results show that older individuals with highly educated children are more likely to engage in health-benefiting behaviors, such as reducing smoking, napping, seeking medical attention, and using clean water sources for cooking. Further analysis reveals substantial heterogeneity across parental gender, age, urban-rural residence, and offspring gender, indicating that the effects are not uniformly distributed across population subgroups. The findings highlight the importance of intergenerational linkages within the family and suggest that children's education may shape parental health behaviors through intra-household differences in access to resources and support. The article presents actionable pathways for enhancing the health behaviors of older demographics and aligns with policy imperatives.

Keywords: Education; health behavior; intergenerational influence

1. Introduction

A large body of research in the social sciences documents a strong positive association between education and health. Education and health, as core components of human capital, exhibit a close and often causal relationship (Xie & Mo, 2014). Empirical evidence consistently shows that individuals with higher educational attainment enjoy better health outcomes, even after controlling for socioeconomic status and demographic characteristics (Baker et al., 2011). Education is therefore widely regarded as a fundamental social determinant of health (Robert and Benedict, 2015). Beyond general health status, scholars have examined whether education shapes specific health behaviors. Individuals with more schooling are less likely to smoke or drink excessively and are more inclined to adopt preventive health behaviors (Cutler & Lleras-Muney, 2010). However, the causal effect of education on health behaviors remains debated. Exploiting a natural experiment in the United Kingdom, Braakmann (2011) finds that additional schooling does not necessarily translate into measurable differences in health-related behavior within cohorts. These mixed findings suggest that the mechanisms linking education and health behaviors remain incompletely understood, particularly in the presence of unequal access to resources

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and information that shape individuals' ability to translate education into health-related actions.

While most of the existing literature focuses on the effect of individuals' own education, a related but less explored question is whether education generates intergenerational spillovers within families and thereby influences the distribution of health-related resources. The downward transmission of human capital from parents to children has been extensively documented, yet potential upward spillovers from children to parents remain understudied. This issue is especially relevant in aging societies such as China, where the share of individuals aged 60 and above is projected to exceed 20 percent of the population and where disparities in access to healthcare, information, and public services remain pronounced. In such a context, families serve as an important unit for reallocating resources across generations. Adult children with higher educational attainment may influence their parents' health behaviors through multiple channels, including the transmission of health knowledge, the provision of financial support, and changes in intra-household decision-making. These channels may be particularly important for individuals facing tighter resource constraints, suggesting that intergenerational spillovers could play a compensatory role in mitigating health-related inequalities. Nevertheless, credible empirical evidence on whether children's education causally affects the health behaviors of middle-aged and elderly parents remains limited.

This paper examines whether children's educational attainment affects the health behaviors of middle-aged and elderly parents, with a particular focus on the role of intra-household resource allocation. By doing so, it aims to shed light on whether education generates broader intergenerational effects that extend beyond the directly educated individual and contribute to the redistribution of health-related resources within families.

This paper contributes to the literature in several respects. First, it extends the literature on intergenerational spillovers by focusing on health behaviors as outcome variables, rather than traditional outcomes such as mortality or subjective well-being. Second, it highlights the role of intra-household resource allocation as a key mechanism linking children's education to parental outcomes, thereby enriching the understanding of how education generates broader family-level effects. Third, by examining multiple behavioral dimensions, this paper provides a more comprehensive account of how intergenerational factors shape health-related decision-making among middle-aged and elderly populations. Finally, the findings offer policy-relevant insights into how family-based channels may complement formal institutions in promoting healthy aging.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and develops the theoretical framework. Section 3 presents the institutional background. Section 4 introduces the data and empirical strategy. Section 5 presents the empirical results. Section 6 concludes.

2. Literature Review

2.1 Socioeconomic Determinants of Health Behaviors

Existing literature has extensively examined the determinants of health behaviors among middle-aged and elderly populations, emphasizing that such behaviors are systematically shaped by unequal access to economic, informational, and social resources.

A first strand of research focuses on individual-level heterogeneity. Psychological factors such as intentions and perceived severity have been shown to influence health-related decision-making (Conner & Norman, 2015). Beyond psychological determinants, socioeconomic inequality plays a central role in shaping behavioral disparities. Individuals with lower socioeconomic status are more likely to engage in smoking and excessive alcohol consumption (West, 2017; Conner & Norman, 2017), reflecting constraints in both material resources and health-related information. Education, as a core component of human capital, is widely recognized as an important determinant of health behaviors, as it influences individuals' capacity to make health-related decisions (Cutler & Lleras-Muney, 2010; Braakmann, 2011). Gender and genetic endowments further contribute to behavioral heterogeneity (Conner & Norman, 2017; Vink et al., 2005).

A second strand of literature emphasizes the role of social and environmental contexts in shaping health behaviors. Social interactions influence behaviors such as smoking (Li and Gilleskie, 2021). At a broader level, environmental factors, which include support from friends and family (West, 2017), community characteristics, work environments, and institutional settings, generate systematic differences in behavioral opportunities. Research conducted by Glass and McAtee (2006) further shows that individual behaviors such as diet, physical activity, and smoking are shaped by the social environment, including factors such as local cigarette taxes, poverty levels, and food availability. This literature suggests that differences in health behaviors are closely associated with underlying patterns of resource allocation.

Beyond lifestyle behaviors, healthcare utilization is also shaped by unequal access to institutional and financial resources. Social capital factors such as special trust, religious beliefs and social participation significantly affect medical service utilization among rural residents in China (Zhou et al., 2022). Institutional arrangements such as insurance systems also play a redistributive role; for example, long-term care insurance reduces hospitalization duration and expeditions among the elderly (Feng, 2020), and insurance coverage significantly affects healthcare utilization decisions (Anderson et al., 2012).

This literature suggests that health behaviors are fundamentally determined by unequal distribution of economic resources, information, and institutional access. However, most existing studies implicitly treat individuals as isolated decision-makers, thereby overlooking an important locus of inequality: the household as a unit of resource allocation and redistribution.

2.2 Family Context and Intergenerational Spillovers

A growing body of literature recognizes that health outcomes and related behaviors are shaped within families through intergenerational interactions. From an economic perspective, the household functions as an internal allocation mechanism through which resources are redistributed across generations.

One strand of literature highlights intergenerational relationships as a channel of non-market resource exchange. Mutual support and high-quality parent-child relationships are associated with improved mental health among older adults (Zheng et al., 2022), while sleep-related health outcomes are jointly determined within families (Wang et al., 2020). These findings suggest that emotional and behavioral resources are not individually determined but are jointly produced within households.

Another strand of research focuses on family structure as a determinant of resource allocation. The number of children is found to have a nonlinear relationship with parental mortality (Tamakoshi et al., 2011; Einiö et al., 2016), indicating that caregiving, financial transfers, and time resources are distributed across offspring in a non-random manner. Similarly, children's gender composition affects parental well-being and health outcomes (Li, 2021; Næss et al., 2017; Chen & Sun, 2022), reflecting heterogeneity in intra-household bargaining power and allocation of family resources.

A particularly important dimension is children's human capital accumulation, especially education. Existing studies show that children's education can affect parental quality of life (Shi, 2016), cognitive functioning, survival expectations (Ma, 2019), and mortality risk (Neve & Fink, 2018). These findings suggest that educated children are able to convert their human capital into tangible and intangible resources for their parents, including information, financial transfers, and institutional access.

However, two gaps remain in the literature. First, most studies focus on parental outcomes such as mortality, cognition, or subjective well-being, while paying limited attention to health behaviors as an important dimension of parental outcomes. Second, existing research has not sufficiently explored how children's education reshapes inequality in access to health-relevant resources within households, particularly in aging societies where family-based support remains central.

This paper addresses these gaps by explicitly framing children as agents of intra-household resource reallocation, through which educational attainment may translate into changes in parental health behaviors.

2.3 Education and Intra-household Allocation

Health behaviors are among the most important determinants of long-term health outcomes. According to the World Health Organization(WHO)'s *Top 10 Threats to Global Health in 2019*, more than 70% of global mortality is attributable to behavioral risk factors such as smoking, alcohol consumption, physical inactivity, and unhealthy diet. These behaviors are shaped not only by individual preferences but also by constraints in information, financial resources, and institutional access.

Existing research has increasingly emphasized that such constraints are often shaped within the family. Adult children, in particular, may influence parental outcomes through multiple channels. Empirical evidence shows that children affect parental life satisfaction and health outcomes through financial support, daily care, and emotional interaction (Lian et al., 2015; Yang et al., 2019), while improvements in living environments also play an important role (Zhuang, 2022). These findings suggest that health-related resources are not allocated at the individual level alone, but are partly determined within households.

This paper builds on this literature by viewing children's education as a key determinant of intra-household resource allocation. Prior studies have shown that children's education is associated with parental cognitive outcomes, survival expectations, and broader measures of well-being (Shi, 2016; Ma, 2019; Neve & Fink, 2018). More educated children tend to have higher earnings, better access to information, and stronger ability to navigate formal institutions, which may translate into multiple forms of support for parents.

From this perspective, parental health behaviors can be understood as the outcome of a broader process of resource allocation within households. In particular, children's education may simultaneously relax informational, financial, and institutional constraints faced by older adults, for example by facilitating participation in medical insurance, improving access to digital information, and enhancing household economic capacity. Examining these mechanisms jointly may therefore provide a more complete account of how intergenerational factors shape health-related decision-making.

3. Institutional Background

China's compulsory education system is governed by the Compulsory Education Law, first enacted in 1986 and subsequently strengthened through a series of reforms. The law mandates nine years of compulsory education, consisting of six years of primary schooling and three years of junior secondary schooling. During this period, public education is largely provided at low or no cost, and school-age children are legally required to enroll and complete the basic education cycle.

The implementation of compulsory education in China has been a gradual but nationwide process, characterized by strong central coordination. Local governments are responsible for enforcement and resource allocation, but the overall institutional framework is set at the national level. Importantly, the timing and design of the reform were not targeted at specific households or regions based on health behaviors, family characteristics, or parental outcomes. Instead, the policy was designed as a universal expansion of basic education coverage.

A key feature of the compulsory education reform is its role in shaping educational attainment across cohorts. By reducing direct and indirect costs of schooling, the policy significantly increased enrollment rates and completed years of education among school-age children. As a result, exposure to compulsory education provides a source of exogenous variation in children's educational attainment that is widely used in the literature (e.g., Ma, 2019).

In this paper, the compulsory education reform serves as an instrumental variable for children's education. The identifying assumption is that the reform affects parental health behaviors only through its impact on children's educational attainment, rather than through any direct channel. Given the institutional design of the policy and its universal implementation, it provides a plausible source of exogenous variation for estimating intergenerational spillover effects.

4. Empirical Models, data and variables

4.1 Data sources

The data used in this article come from the China Family Panel Studies (CFPS) database. Conducted by the Institute of Social Science Survey at Peking University, CFPS is a national, large-scale and multidisciplinary social tracking survey project. The CFPS sample covers 25 provinces/municipalities/autonomous regions, with a target sample size of 16,000 households, and the survey targets include all family members in the sampled households. The CFPS conducted pilot surveys in Beijing, Shanghai and Guangdong in 2008 and 2009, respectively, and officially launched the survey in 2010. All household members defined in the 2010 baseline survey and their future biological/adopted children are considered genetic members of the CFPS and become permanent tracking subjects. This article uses mixed cross-sectional data from the CFPS in 2010, 2012, 2014, 2016 and 2018, focusing on parents aged 45-74, corresponding to the World Health Organization's 'middle-aged' and 'young-old' groups. Among them, the parents of 4,710 children were interviewed five times consecutively, accounting for about 18.42% of the total; 6,176 were interviewed four times, accounting for about 22.84%; 6,120 were interviewed three times, accounting for about 22.63%; 5,878 were interviewed twice, accounting for about 21.74%; and the remaining 4,259 parents of children were interviewed only once, accounting for about 15.38% of the total. In subsequent robustness tests, the upper age limit of the parent generation is extended. The age of the offspring is controlled at 22 and above, and to avoid the influence of extreme values in the data, only samples with an age difference of 18 years or more between parents and offspring are included in this article.

4.2 Empirical model and variables

This article uses mixed cross-sectional data to test the effect of children's educational attainment on parents' health behavior, setting up the following linear probability model (LPM):

$$Behavior_i = \alpha_0 + \beta_0 \cdot ChildEduy_i + \gamma_0 \cdot Controls_i + {\mu_0}_i \qquad \textit{(Eq.1)}$$

In this article, *Behavior* refers to the health behaviors of parents and served as explained variable. It is measured by five aspects: smoking, drinking, sleeping, medical visits and using water for cooking. *Behavior* is a dummy variable where a value of 1 indicates that the parents engage in a particular health behavior and a value of 0 that they do not. *ChildEduy* represents the education level of the children, measured by the number of years of education they have completed, and is the primary explanatory variable in the analysis. Given that parents may have multiple children, we select the child with the longest educational duration among all siblings to represent the educational status of the family. This approach ensures a consistent and meaningful correspondence between each parent and their child's educational achievement.

The coefficient β_0 is the main focus of this study. In the regression model, *Controls* refers to a set of control variables, including those at the individual-level characteristics of both parents and children, as well as variables at the family and provincial levels. The term μ_0 denotes the random error term in the model. Descriptive statistics for all the variables used in the analysis are presented in Table 1.

4.2.1. Parents' Health Behaviors

The explanatory variables in this paper are indicators of the parents' generation, measured across five dimensions: smoking, drinking, sleep, seeking medical attention, and the water used for cooking.

Smoking (*Smoke*)

This variable is represented by a binary indicator reflecting "whether the individual has smoked cigarettes in the past month." A value of 1 denotes that the individual smoked in the past month, whereas a value of 0 indicates that they did not.

Drinking (*Drink*)

This variable is captured by a binary indicator denoting "whether the individual has drunk alcohol more than three times a week in the past month." A value of 1 is assigned if the individual drank alcohol more than three times weekly, and 0 if otherwise.

Sleep (*Siesta*)

This variable is measured by a binary indicator reflecting "whether the individual has the habit of taking afternoon naps." A value of 1 indicates that the individual takes a daily nap, while a value of 0 indicates they do not.

Table 1 Descriptive Statistics of Variables

	Variable Name	Description	Observations	Mean (S.D.)	Min	Max
Parents' health behaviors	Smoke	Smoking in the past month =1, 0 otherwise	41,731	0.303 (0.459)	0	1
	Drink	Drinking more than 3 times a week in the past month = 1, 0 otherwise	41,730	0.171 (0.376)	0	1
	Siesta	Siesta = 1, 0 otherwise	41,727	0.540 (0.498)	0	1
	Doctor	Prompt medical care after discomfort = 1, 0 otherwise	15,107	0.783 (0.412)	0	1
	Cleanwater	Cooking water is tap/mineral/purified/filtered = 1, 0 otherwise	44,118	0.657 (0.475)	0	1
Core explanatory variables	ChildEduy	Children's years of schooling	41,233	10.83 (4.071)	0	23
control variables	ChildGender	Children's gender (Male =1, Female =0)	42,934	0.726 (0.446)	0	1
	ChildAge	Children's age	44,218	30.96 (6.956)	22	56
	ChildNumber	Children' number	44,106	2.212 (1.098)	0	9
	gender	Parents' gender (Male =1, Female =0)	43,464	0.475 (0.499)	0	1
	age	Parents' age	44,223	57.26 (7.780)	45	74
	Marriage	Parents' marital status (values 1-5 representing unmarried, married (with spouse), cohabiting, divorced, and widowed respectively)	43,463	2.296 (0.884)	1	5
	eduy	Parents' years of schooling	42,948	5.660 (4.537)	0	19
	log(perincome)	The logarithm form of household per capita income (in yuan)	42,946	9.023 (1.200)	-1.099	14.23
	urban	Urban-rural classification variable (rural = 0, urban = 1)	43,678	0.436 (0.496)	0	1

Seeking Medical Attention (*Doctor*)

This variable is assessed based on whether the parents sought medical attention promptly when having felt physically uncomfortable during the past two weeks. A value of 1 is assigned if they see a doctor, and 0 if they did not.

Water Used for Cooking (*Cleanwater*)

This variable is represented by a binary indicator denoting "whether the water used for cooking is tap water, mineral water, purified water, or filtered water." A value of 1 is assigned if any of these water types are used, and 0 if otherwise.

These indicators collectively capture the health behaviors of the parents, providing a comprehensive measure for the analysis.

4.2.2. The Educational Level of the Children

In this study, the children's years of schooling is utilized as the primary explanatory variable. In order to measure the educational attainment of children more accurately, we set the minimum age limit of 22 years, ensuring that the subjects have typically completed their undergraduate studies, thereby avoiding any potential bias from incomplete education.

The duration of education for this representative child is employed as the core explanatory variable for the regression analysis. As illustrated in Table 1, there is a notable disparity in educational levels between the children and their middle-aged and elderly parents. Specifically, the average educational duration for the parental generation is only 5.7 years, while their children's average educational duration extends to 10.8 years. This significant difference indicates that the overall educational status of the offspring generation is substantially higher than that of their parents.

Moreover, since the educational status of offspring may be influenced by unobservable factors that simultaneously affect both offspring education and parental health behaviors, such as genetic traits, and because parental health behaviors may impact the investment in offspring education by affecting the parents' own educational levels, addressing endogeneity is crucial. To mitigate these endogenous issues, we adopt the methodology used by Ma (2019) and Liu (2022) to set the variable *Exposure*, which denotes the exposure of children to the nine-year compulsory education law in China, as an instrumental variable (IV) for children's years of schooling. *Exposure* primarily measures the extent to which children are influenced by compulsory education. If children have a longer exposure to the compulsory education, the potential impact of the compulsory education law will be greater, leading to a higher educational level among the children.

The instrumental variable method is specified as shown in Equation 2, with the index ranging between 0 and 1. The distribution for each stage is presented in Table 2.

$$Exposure = \begin{cases} 0 & , if \ \ BirthYear < FirstCohort; \\ \frac{BirthYear - FirstCohort + 1}{10} & , \ if \ \ FirstCohort \leq BirthYear \leq FullCohort \\ 1 & , if \ \ BirthYear > FullCohort. \end{cases} \qquad (Eq.2)$$

The variable *BirthYear* denotes the birth year of the children. *FirstCohort* represents the first birth cohort eligible for compulsory education, with individuals in this cohort being exactly 15 years old when the Compulsory Education Law(CEL) is implemented. Conversely, *FullCohort* indicates the birth cohort eligible for the entire nine-year compulsory education period, with individuals in this cohort being exactly 6 years old at the law's implementation.

The relationship between a child's age at the time of the Compulsory Education Law's implementation and the instrumental variable *Exposure* is illustrated in Fig.2. An *Exposure* value of 0 implies that the individual exceeded the age limit for compulsory education by the enactment year and is thus entirely ineligible for it. In contrast, an *Exposure* value of 1 indicates that the individual has not yet reached the starting age for compulsory education by the enactment year, making them eligible for the complete nine-year compulsory education.

Table 2 Distribution of the Instrumental Variable *Exposure*

Exposure	Frequency	Percent
0	3,993	16.72
0.1-0.9	10,607	23.98
1	29,623	66.99
Total	41,808	100.00

4.2.3. Control variables

The control variables in this paper encompass four primary dimensions: parental, offspring, family, and provincial levels. In particular, the parental-level control variables include personal characteristics such as gender, age, age squared, education level, household registration status, marital status, and number of children. The offspring-level control variables account for the gender and age of the children. At the family level, the control variable is household per capita income, adjusted for inflation using 2010 as the base year.

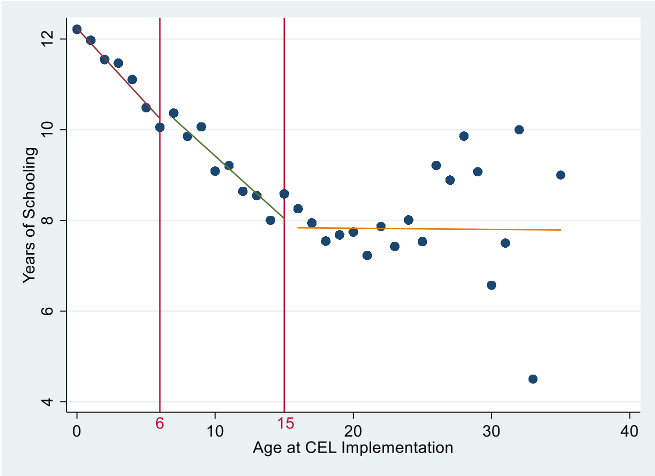

Fig.1 Adult children's years of schooling by age at CEL implementation

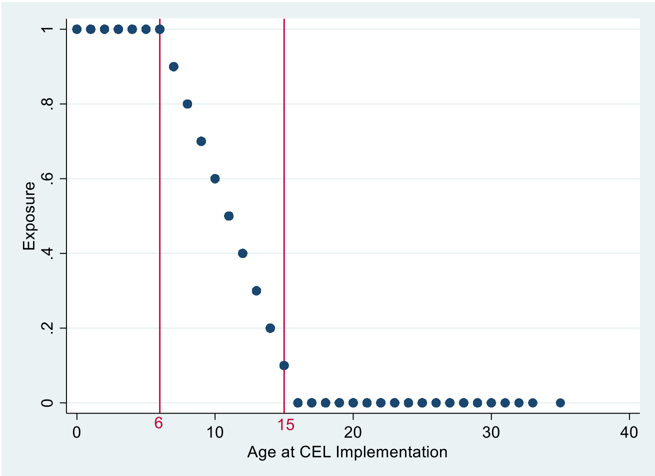

Fig.2 Constructed exposure to the CEL

5. Empirical Results

5.1 Baseline Regression

This section presents baseline regression results based on the linear probability model (LPM) specified in Equation 1, estimated using ordinary least squares (OLS). The results are shown in Table 3. The findings indicate that the adult children's education significantly and negatively affects the smoking behavior of middle-aged and elderly parents at the 1% significance level. Specifically, for every additional year of education completed by children, the probability of their parents smoking decreases by an average of 0.0053. However, the impact of children's education on parental drinking behavior is not significant. This could be due to the social acceptance of moderate drinking, while the explanatory variable in this study focuses on whether parents drink more than three times a week, rather than the quantity or pattern of alcohol consumption. In addition, an increase in children's education level significantly enhances the likelihood of middle-aged and elderly parents taking afternoon naps and seeking medical attention promptly when experiencing physical discomfort. For every additional year of education, the probability of parents seeking medical attention promptly upon feeling unwell increases significantly by an average of 0.0042, and the probability of taking afternoon naps increases significantly by an average of 0.0056. Column (5) in Table 3 shows that the coefficient of *ChildEduy* is significantly positive at the 1% level, suggesting that for every additional year of education completed by children, the probability of their parents using clean water sources for cooking increases significantly by an average of 0.0101. In summary, the higher educational levels in children are associated with an increased likelihood of middle-aged and elderly parents taking afternoon naps, using clean water sources for cooking, seeking medical attention promptly when feeling unwell, and reducing smoking behavior.

The results also show that an increase in the number of adult children significantly increases the probability of smoking among middle-aged and elderly parents while decreasing the probability of them taking afternoon naps and using clean water sources for cooking. In addition to the number of children, the study also found significant impacts of the gender of children on the health behaviors of older parents. As shown in Column (3) of Table A1 in Appendix I, the gender of the children has a significantly negative effect on the probability of older parents seeking medical attention promptly upon feeling unwell, at the 1% significance level. Specifically, middle-aged and elderly individuals with daughters are more likely to seek medical attention promptly when experiencing physical discomfort compared to those with sons. Furthermore, the results in Column (5) of Table A1 in Appendix I indicate that daughters are more likely than sons to encourage their parents to use clean water sources for cooking. This finding aligns with previous research that suggests daughters tend to contribute more to the health and well-being of elderly parents, indicating that daughters are more conducive to promoting health behaviors among their parents.

To further address potential endogeneity issues, this study constructs *Exposure* as an IV for children's educational level in the regression analysis. To examine the validity of the IV, a first-stage regression equation, as shown in Equation 3, is formulated. The regression results are presented in Table 4.

$$ChildEduy_i = \delta_0 + \delta_1 \cdot Exposure_i + \delta_2 \cdot Controls_i + u_i \qquad (Eq.3)$$

The first-stage regression results, presented in Table 4, reveal that the estimated coefficients for the IV *Exposure* are significantly positive at the 1% significance level. This finding indicates a strong positive correlation between the prescribed number of years of compulsory education and the actual educational attainment of children, thereby affirming the exogeneity of the IV. Furthermore, the minimum eigenvalue statistic reported in Table 5 substantiates that there is no concern regarding weak instrumental variables in this model.

Table 3 The OLS regression results for Eq.1

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Smoke	Drink	Doctor	Siesta	Cleanwater
ChildEduy	-0.0053***	-0.0010	0.0042***	0.0056***	0.0101***
	(0.0008)	(0.0007)	(0.0012)	(0.0011)	(0.0013)
Control variables	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family
Observations	36,485	36,484	13,060	36,480	(0.2062)

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The reported results in the table are coefficients corresponding to the explanatory variables. For example, the number -0.0053 corresponding to *ChildEduy* in (1) indicates that, with other variables remaining unchanged, for each additional year of education for children, the probability of their parents smoking decreases by an average of 0.0053.

This study uses a two-stage least squares (2SLS) approach to address potential endogeneity concerns in the estimation. The results, presented in Table 5, demonstrate that the findings obtained through the IV are consistent with those from the OLS regression, but with magnified effects of children's education on the health behaviors of older parents. Specifically, children's education significantly reduces the probability of smoking among middle-aged and elderly parents at the 5% significance level. Each additional year of children's education decreases the average probability of smoking among these parents from 0.0053 to 0.022. In addition, while children's education continues to inhibit parents' drinking behavior, the results remain statistically insignificant after employing the IV. Regarding healthcare-seeking behavior, the findings indicate that higher levels of children's education significantly increase the likelihood that older parents will seek medical attention promptly upon feeling unwell. Each additional year of children's education raises this probability by 0.080 at the 1% significance level. Similarly, the probability of parents taking afternoon naps also rises with their children's educational attainment, suggesting an impact on their sleep patterns. Each additional year of children's education increases the likelihood of parents taking afternoon naps by 0.018. Finally, in terms of the use of clean water sources for cooking, the results in Table 5 suggest that children's education significantly enhances the probability of parents using clean water sources. Each additional year of children's education raises this probability by 0.064.

Table 4 Table of Results of First-stage Regression Using IV

VARIABLES	(1) Smoke ChildEduy	(2) Drink ChildEduy	(3) Doctor ChildEduy	(4) Siesta ChildEduy	(5) Cleanwater ChildEduy
Exposure	1.862***	1.863***	1.720***	1.862***	1.816***
	(0.201)	(0.201)	(0.260)	(0.201)	(0.199)
Control variables	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family
Observations	36,485	36,484	13,060	36,480	37,598

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively.

Table 5 Table of IV-2SLS Regression Results for Eq.1

VARIABLES	(1) IV-2SLS Smoke	(2) IV-2SLS Drink	(3) IV-2SLS Doctor	(4) IV-2SLS Siesta	(5) IV-2SLS Cleanwater
ChildEduy	-0.0219**	-0.0002	0.0796***	0.0182*	0.0642***
	(0.0086)	(0.0073)	(0.0171)	(0.0109)	(0.0141)
Control variables	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family
Minimum eigen-value	313.343	313.728	101.719	313.517	306.94
Observations	36,485	36,484	13,060	36,480	37,598

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The reported results in the table are coefficients corresponding to the explanatory variables. For example, the number -0.0219 corresponding to *ChildEduy* in (1) indicates that, with other variables remaining unchanged, for each additional year of education for children, the probability of their parents smoking decreases by an average of 0.0219. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.

5.2 Heterogeneity Analysis

To further validate Hypothesis 5, this section will perform a heterogeneity analysis of the aforementioned five health behaviors of middle-aged and elderly parents across four dimensions: parental gender, urban-rural residence, age, and offspring gender. For age, this study follows the WHO's definitions, categorizing parents aged 45-59 as "middle-aged" and those aged 60-74 as "younger elderly". Regarding parental gender, the analysis will separately examine the impact of offspring's educational level on the health behaviors of fathers and mothers. Concerning offspring gender, the study will explore the influence of daughters and sons on the health behaviors of middle-aged and elderly parents. Lastly, in terms of urban-rural residence, the analysis will utilize the urban-rural classification variables from the National Bureau of Statistics to distinguish between rural and urban areas, conducting grouped regression analyses and testing for differences in coefficients between groups using seemingly unrelated regression(SUR) models.

5.2.1 Smoke

The regression results using the IV are presented in Fig.3 (see more details in Appendix I Table A2). An increase in children's educational level reduces the probability of smoking in both the "middle-aged" (45-59 years old) and "younger elderly" (60-74 years old) parents, but neither effect is statistically significant. The SUR test's P-value for the coefficient difference is greater than 0.1, indicating no heterogeneity based on parents' age.

In contrast, heterogeneity emerges with respect to parental gender. An increase in children's education significantly decreases fathers' smoking, but not mothers'. The difference in coefficients is statistically significant at the 5% level. This indicates that behavioral responses to children's education are unevenly distributed across gender groups, with fathers being more responsive in terms of smoking behavior. One possible explanation is that baseline differences in smoking prevalence may translate into differences in behavioral adjustability across groups. This pattern is consistent with previous evidence showing that higher education in adult children significantly improves the survival probability of fathers but has no significant effect on mothers' survival probability (Cui et al., 2021). These findings suggest that the impact of children's education may operate differently across parental subgroups, potentially through heterogeneous pathways that will be further explored in the mechanism analysis.

Regarding urban-rural differences, children's education significantly reduces smoking among rural parents but not among urban parents. However, the p-value for the coefficient difference is 0.420, indicating no statistically significant heterogeneity. This suggests that while point estimates differ, the responsiveness of smoking behavior to children's education does not vary systematically by residential status in a statistically meaningful way. Any observed differences may therefore reflect baseline environmental conditions rather than structural differences in responsiveness.

For offspring gender, increased education of both daughters and sons significantly reduces parental smoking. Each additional year of a daughter's education reduces the probability of parents smoking by 0.035, while each additional year of a son's education reduces it by 0.012. The p-value for the coefficient difference is greater than 0.1, indicating no statistically significant heterogeneity by offspring gender. This implies that the association between children's education and parental smoking is broadly consistent across sons and daughters, despite differences in point estimates.

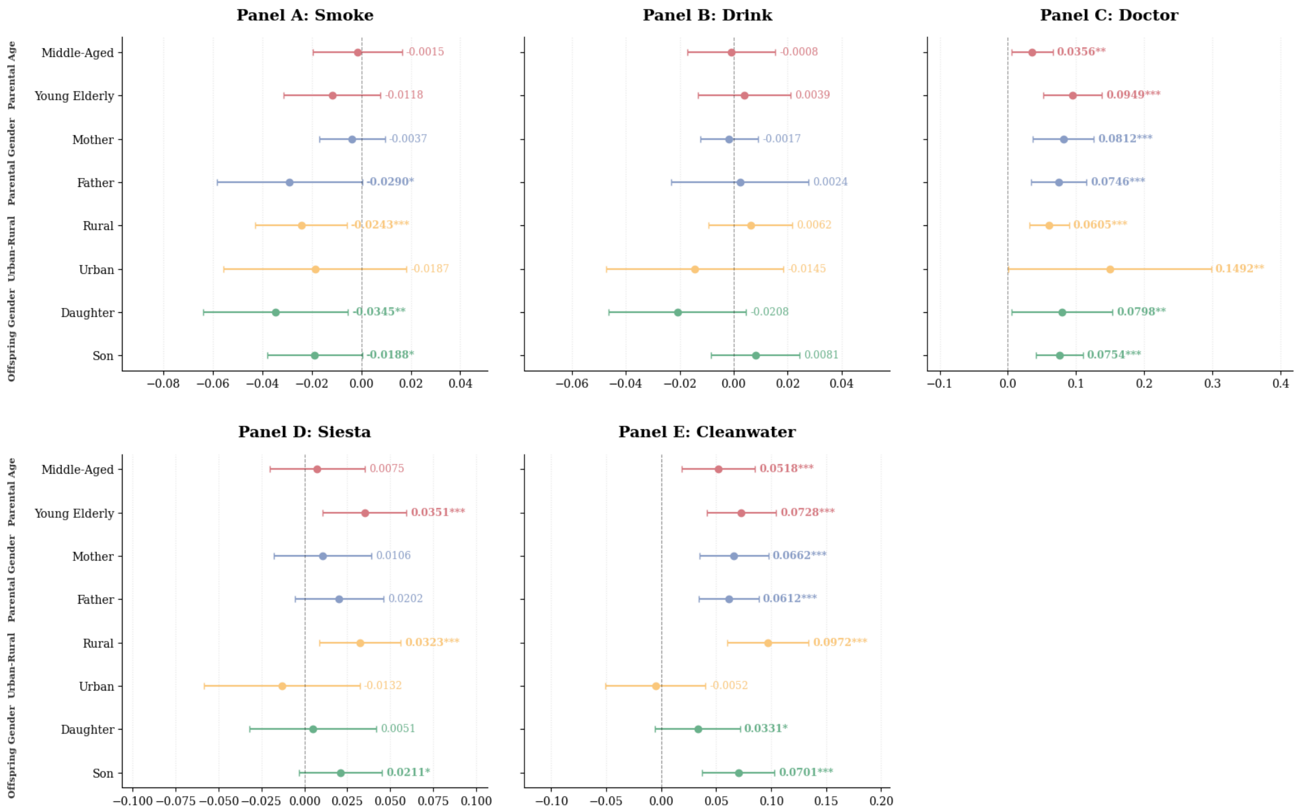

Fig.3 The Heterogeneous Regression Results for Parents' Health Behaviors

5.2.2 Drink

The regression results using IV are in Fig.3 (see more details in Appendix I Table A3). For parental age, increased children's education reduces drinking probability in the "middle-aged" group and increases it in the "young elderly" group, though neither effect is statistically significant. Previous studies have indicated that alcohol consumption affects health differently across age groups. Younger elderly individuals have a higher safe drinking level, possibly explaining their higher drinking frequency. The P-value for the coefficient difference is greater than 0.1, indicating no significant heterogeneity based on parental age.

For parental gender, increased children's education reduces mothers' drinking probability and increases fathers', though neither effect is significant. Mothers are more likely to heed their children's advice, explaining this difference. The P-value for the gender coefficient difference is 0.290, indicating no significant heterogeneity based on parental gender.

For urban-rural differences, Table A3 shows that increased children's education raises rural parents' drinking probability and lowers urban parents', though neither effect is significant. The P-value for the urban-rural coefficient difference is 0.030, indicating heterogeneity at the 5% significance level. Children's education has a greater inhibitory effect on urban parents' drinking. Such differences may reflect disparities in baseline behavioral environments and constraints faced by different populations.

For offspring gender, increased daughters' education reduces parents' drinking probability, while increased sons' education increases it, though neither effect is significant. The P-value for the gender coefficient difference is 0.020, indicating heterogeneity at the 5% significance level. Daughters' education more effectively reduces parents' drinking probability than sons'.

5.2.3 Doctor

The regression results using IV are in Fig.3 (see more details in Appendix I Table A4). Regarding parents' age, improved children's education significantly boosts the medical-seeking behavior of both "middle-aged" and "young elderly" groups. The SUR test shows a P-value of 0.030, indicating significant differences at the 5% level. Children's education has a greater impact on timely medical attention for young elderly individuals, likely due to their lower mobility and awareness.

For parents' gender, increased children's education positively affects both fathers and mothers seeking timely medical attention. The P-value for the coefficient difference is 0.440, showing no significant difference, indicating no heterogeneity based on parental gender.

In urban-rural terms, increased children's education significantly enhances medical-seeking behavior for both rural and urban parents. The P-value of 0.000 indicates significant heterogeneity at the 1% level. Children's education has a more substantial effect on urban parents, likely due to better medical resources and access in urban areas. Rural parents often lack nearby children for practical assistance, relying more on verbal encouragement.

For offspring's gender, improved education levels of both daughters and sons significantly increase the probability of parents seeking timely medical attention. Each additional year of daughters' education raises the probability by 0.080, and each year of sons' education raises it by 0.075. The SUR test indicates no significant heterogeneity in the impact of children's education based on offspring gender.

5.2.4 Siesta

The regression results using IV are presented in Fig.3 (see more details in Appendix I Table A5). For the parental age, longer years of children's education increase the probability of midday naps in both the "middle-aged" and "young elderly" groups, with a more significant impact on the young elderly. The SUR test indicates a P-value of 0.060, showing a difference in coefficients at the 10% significance level. This suggests children's education has a greater impact on midday naps for the young elderly, likely due to their need to supplement sleep.

Regarding the parents' gender, increased children's education positively affects siesta behavior for both fathers and mothers. The SUR test shows a P-value of 0.350, indicating no significant difference between genders. This suggests no heterogeneity in the impact of offspring's education on siesta based on parental gender.

For urban-rural differences, increased children's education significantly boosts midday nap behavior in rural parents but has a negative, insignificant effect on urban parents. The P-value of 0.000 indicates a significant difference at the 1% level, suggesting a greater impact on rural parents. According to Lu and Wang (2022), healthy aging indicators exhibit significant age and cohort effects, with rural elderly individuals being at a disadvantage in terms of health. Due to their declining health indicators, rural elderly parents are more likely to be influenced by their children's education to engage in health-beneficial behaviors.

Regarding offspring gender, increased sons' education significantly raises the probability of midday naps for middle-aged and elderly parents. Each additional year of sons' education increases this probability by 0.02. In contrast, daughters' education has no significant impact. The P-value for the difference in coefficients is greater than 0.1, indicating no heterogeneity in the impact of children's education on midday naps based on offspring gender.

5.2.5 Cleanwater

The regression results using IV are in Fig.3 (see more details in Appendix I Table A6). For parental age, longer children's education significantly increases the use of clean water for cooking. Each additional year of children's education raises this probability by 0.052 for middle-aged parents and 0.073 for young elderly parents. The SUR test shows a P-value above 0.1, indicating no significant age-related heterogeneity in this effect.

Regarding the parental gender, increased children's education significantly enhances the use of clean water for cooking by both fathers and mothers. The SUR test's P-value of 0.390 indicates no significant gender-based heterogeneity.

For the urban-rural differences, increased children's education significantly boosts the use of clean water for cooking among rural parents but has an insignificant negative effect on urban parents. The P-value of 0.000 shows significant heterogeneity. In China, the urban water quality assurance rate is higher than that in rural areas, thus the impact of children's education on improving the use of clean water sources for urban parents is minimal. However, for rural parents, due to limitations in rural conditions, the availability and convenience of accessing clean water sources are lower, leaving more room for improvement. Therefore, the improvement in children's education has a greater impact on the use of clean water sources by rural parents.

For offspring gender, higher education levels of both daughters and sons significantly increase the use of clean water for cooking by older parents. The P-value of 0.020 indicates significant heterogeneity based on offspring gender. Each additional year of sons' education raises this probability by 0.07, which is higher than the impact of daughters' education.

5.3 Mechanism Analysis

The influence of children's characteristics on their parents is multifaceted. This article posits several factors, including parents' medical insurance participation, internet use, and offspring's income. It examines how children's education affects parental health behaviors through an upward spillover process operating within the family.

5.3.1 Health Insurance Participation Status

To examine whether children's education level influences their parents' health behaviors through their parents' participation in medical insurance, this article constructs a model as shown in Equation 4.

$$Insurance_i = \alpha_1 + \beta_1 \cdot ChildEduy_i + \gamma_1 \cdot Controls_i + {\mu_1}_i \qquad (Eq.4)$$

In this model, *Insurance* is a parental-level dummy variable, indicating whether they have medical insurance or not. The types include public, urban employee, urban resident (including coverage for the elderly and children), supplementary, and the new rural cooperative medical insurance. If the parents have any, *Insurance* equals 1; otherwise, 0. The other variables are set as Equation 1. The results of the IV-2SLS regression are in Table 6.

The coefficient of *ChildEduy* significantly positively affects parents' medical insurance participation at the 1% level. As children's education rises, they understand medical insurance better and persuade parents to participate, impacting intergenerational behaviors. This confirms that children's education influences parents' health behaviors by promoting insurance participation. From this perspective, insurance participation reflects a key dimension of intra-household resource reallocation, through which educational differences across children are translated into unequal access to health protection among parents.

Moreover, children's gender is significantly associated with parental insurance participation at the 1% level. Households with daughters are less likely to have parental insurance coverage compared to those with sons, consistent with Liang and Chen (2022). This finding suggests that the distribution of intergenerational support within the household is not uniform. In particular, daughters may be more likely to provide informal or direct support, which may substitute for formal insurance coverage, thereby generating differences in how health-related resources are allocated across families.

5.3.2 Internet Access

To assess whether children's education level influences their parents' health behaviors through their parents' internet usage, this article constructs a model as shown in Equation 5.

$$Internet_i = \alpha_2 + \beta_2 \cdot ChildEduy_i + \gamma_2 \cdot Controls_i + \mu_{2i} \qquad (Eq.5)$$

In this model, *Internet* is a dummy variable representing "whether or not the parents use the internet", where "using the internet" refers to accessing the internet through telephone lines, local area networks, wireless networks, and other means. If they use the internet, it's assigned 1; otherwise, 0. The other variables remain consistent with Equation 1. The regression results appear in Table 6.

The coefficient on *ChildEduy* is significantly negative at the 1% level, indicating that higher children's education is associated with lower parental internet use. One interpretation is that more educated children may take over part of the information acquisition process within the household, reducing the need for parents to independently search for health-related information online.

This shift suggests a change in how information is obtained within the family, from individual digital search toward intergenerational transmission. In this sense, internet use may partly capture the degree of reliance on external versus intra-family information channels. At the same time, intensive internet use may be associated with less structured daily routines and potential health-related discomforts. Against this background, the reduction in parental internet use associated with children's education may reflect a reallocation of informational responsibility within the household rather than a direct behavioral restriction.

Overall, the results are consistent with the view that children's education influences parental health behaviors by reshaping information acquisition patterns within the family.

5.3.3 Offspring's Income

To investigate whether children's education level impacts their parents' health behaviors through their children's income, this article constructs a model as shown in Equation 6.

$$Log(ChildIncome)_i = \alpha_3 + \beta_3 \cdot ChildEduy_i + \gamma_3 \cdot Controls_i + \mu_{3_i} \qquad \textit{(Eq.6)}$$

In this model, *Log(ChildIncome)* represents the logarithm of the child's income, and the other variables remain consistent with Equation 1. The results are presented in Table 6.

The coefficient of *ChildEduy* significantly increases children's income at 1%. This result implies an improvement in the financial capacity of children, which may relax budget constraints faced by parents and thereby influence their health-related decisions. better financial conditions within the family can translate into greater ability to access healthcare services and make healthier choices. Existing evidence also suggests that higher levels of intergenerational support from children are associated with better mental health among older adults (Hou et al., 2021), highlighting the broader role of financial and emotional transfers within the family. Children's education affects parental health behaviors through improvements in intra-household financial resources.

Table 6 Table of Mechanism Test Regression Results

	(1)	(2)	(3)
	IV-2SLS	IV-2SLS	IV-2SLS
VARIABLES	Insurance	Internet	Log(ChildIncome)
ChildEduy	0.0569***	-0.0232***	0.5487***
	(0.0083)	(0.0058)	(0.0579)
Control variables	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Cluster	Family	Family	Family
Minimum eigenvalue	303.936	199.776	271.708
Observations	37,393	28,428	22,887

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. A value of 1 for *Insurance* indicates that the parent has medical insurance. A value of 1 for *Internet* indicates that the parent has access to the Internet. *Log(ChildIncome)* is the logarithm of the offspring's income. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.

5.4 Robustness Test

5.4.1 Changing the regression model

To ensure the robustness of our results, this section substitutes Equation 1 with a Logit model, as specified in Equation 7, for regression analysis.

$$P_i = \frac{1}{1+e^{-(\theta_0+\theta_1 \cdot ChildEduy_i+\theta_2 \cdot Controls_i)}}$$

$$L_i = ln\left(\frac{P_i}{1-P_i}\right) = \theta_0+\theta_1 \cdot ChildEduy_i + \theta_2 \cdot Controls_i \qquad (Eq.7)$$

In this Logit model, P_i represents the probability of parents engaging in the specified health behaviors, while the other variables remain consistent with those in Equation 1. Detailed regression outcomes are provided in Table A7 in Appendix I.

The results of the Logit model reveal that an increase in children's educational level significantly decreases the likelihood of their parents smoking and drinking, while notably increasing the likelihood of their parents taking naps, cooking with clean water, and seeking timely medical attention when feeling unwell. As shown in Table A7 , for each additional year of education attained by their children, the probability of parents smoking decreases by 0.53%, the probability of seeking timely medical attention when feeling unwell increases by 0.43%, the probability of taking naps increases by 0.56%, and the probability of cooking with clean water increases by 0.95%. Therefore, even after substituting the regression model, the primary conclusions of this paper remain robust and valid.

5.4.2 Adding control variables

Considering that parents' health behavior choices may be influenced by their own health status, this section incorporates parents' self-reported health status--*selfhealth*, as an additional control variable into the linear probability model in Equation 1 resulting in the model specified in Equation 8.

$$Behaviour_i = \delta_0 + \delta_1 \cdot ChildEduy_i + \delta_2 \cdot selfhealth_i + \delta_3 \cdot Controls_i + u_{1_i} \qquad \textit{(Eq.8)}$$

In this model, *selfhealth* is categorized into five levels, ranging from 1 to 5, representing healthy, fair, relatively unhealthy, unhealthy and extremely unhealthy, respectively. A higher rating indicates poorer self-reported health status. The other variables in Equation 8 remain consistent with those in Equation 1. Using *Exposure* as an IV for children's education, as in Equation 1, detailed regression results are provided in Table A8 in Appendix I.

After accounting for parents' self-reported health, the results indicate that children's educational level continues to significantly reduce the probability of older parents smoking and significantly increases the probability of seeking timely medical attention when feeling unwell. Additionally, it increases the likelihood of older parents taking a siesta and using clean water for cooking. These findings are consistent with the main conclusions previously obtained.

5.4.3 Adjust the sample period

In this section, we adjust the sample period to the years 2010, 2014, and 2018. Regression analysis is conducted using IV-2SLS with data from these three years, utilizing the models specified in Equation 1 and Equation 3. The detailed regression results are presented in Table A9 in Appendix I.

After shortening the sample period, the findings remain consistent: an increase in children's educational level significantly reduces the probability of older parents smoking, and it continues to have a significant positive impact on the likelihood of older parents seeking timely medical attention when feeling unwell, taking naps, and using clean water for cooking. Specifically, for each additional year of education their children receive, the smoking probability of middle-aged and elderly individuals decreases by 0.020, the probability of seeking timely medical attention when feeling unwell increases by 0.086, the probability of taking naps increases by 0.031, and the probability of using clean water for cooking increases by 0.017. Therefore, the main conclusions of this paper remain robust and valid.

5.4.4 Adjust the upper age limit for parents

In this section, we adjust the upper age limit of the parents to 80 years old. Regression analysis is conducted using this modified dataset, employing Equation 1 and Equation 3. The detailed regression results are presented in Table A10 in Appendix I.

After adjusting the upper age limit of parents, it is found that an increase in children's educational level still significantly reduces the probability of smoking among middle-aged and elderly parents.

Additionally, it significantly increases the likelihood of older parents seeking prompt medical attention when feeling unwell. Furthermore, as the educational level of children increases, the probability of older parents taking naps and using clean water for cooking also increases. These findings once again confirm the robustness of the main results obtained in this paper.

5.4.5 Replace the core explanatory variable

We further replace the core explanatory variable with the highest educational level of children, *ChildEdu*, as a proxy variable for their educational attainment. *ChildEdu* is a categorical variable ranging from 1 to 8, representing educational stages from kindergarten through to a doctorate. The specific distribution is detailed in Table 7. The regression model is specified in Equation 9.

$$Behavior_i = \varphi_0 + \varphi_1 \cdot ChildEdu_i + \varphi_2 \cdot Controls_i + u_{2_i} \quad \text{(Eq.9)}$$

Table 7 Distribution of children's highest level of education

ChildEdu	Value	Frequency	Percent
Kindergarten	1	2,578	5.93
Primary School	2	6,471	14.88
Middle school	3	13,652	31.40
High school	4	8,806	20.26
2- or 3-year college	5	6,201	14.26
Bachelor's degree	6	5,297	12.18
Master's degree	7	454	1.04
Doctoral degree	8	16	0.04
Total		43,475	100.00

In Equation 9, *ChildEdu* serves as the key explanatory variable, representing the highest educational qualification of the children, while the other variables are consistent with those in Equation 1. Using *Exposure* as an IV for *ChildEdu*, the IV-2SLS regression results are detailed in Table A11 in Appendix I.

The findings indicate that using the highest educational qualification of children as a proxy for their educational attainment, the main conclusions of this paper remain robust. An increase in children's educational qualification reduces the probability of smoking among middle-aged and elderly parents, while increasing the likelihood of them taking naps, using clean water for cooking, and seeking timely medical attention when feeling unwell.

6. Conclusion

This paper studies whether and how children's education shapes the health behaviors of middle-aged and elderly parents, with a particular focus on the role of intra-household resource allocation and inequality. In the presence of imperfect public provision and uneven access to health-related resources, families remain a central unit through which resources, information, and support are redistributed across generations. Education, as a key determinant of socioeconomic status, may therefore generate not only individual returns but also intergenerational spillovers that reshape the allocation of health-related resources within households.

The empirical results provide consistent evidence that children's education significantly improves a range of parental health behaviors. Specifically, higher educational attainment of children reduces the likelihood that parents smoke, increases the probability that they seek medical attention when ill, and promotes healthier daily habits such as taking naps and using clean water for cooking. These findings suggest that children's education enhances both preventive and health-seeking behaviors of older adults, pointing to a broader role of education in shaping health beyond the individual level.

Importantly, the effects are not uniform across populations, highlighting the interaction between intergenerational spillovers and existing social inequalities. The reduction in smoking is more pronounced among fathers than mothers, suggesting gendered differences in behavioral responsiveness within households. The effect on drinking behavior varies by both urban-rural residence and the gender of children, with stronger effects observed in urban areas and among daughters. In addition, the impact of children's education on medical-seeking behavior is more substantial among the younger elderly, while improvements in daily health practices, such as taking naps and using clean water, are more evident among rural parents. These heterogeneous patterns indicate that the influence of children's education is shaped by baseline disparities in access to resources, information, and health infrastructure, and that intergenerational spillovers may play a compensatory role in more resource-constrained settings.

To further understand the underlying mechanisms, the analysis examines several channels through which children's education may affect parental health behaviors. The results suggest that children's educational attainment operates through increased parental participation in medical insurance, improved access to information via the internet, and higher income of offspring. These mechanisms point to a common pathway: more educated children relax household-level resource constraints and enhance information flows, thereby enabling parents to adopt healthier behaviors. In this sense, children's education contributes to a reallocation of economic and informational resources within the family, partially mitigating inequalities faced by older adults.

The findings highlight that education has important intergenerational externalities operating through the family. In contexts where public support systems are incomplete and access to health-related resources is uneven, the education of children can serve as an informal mechanism that redistributes both economic and informational resources, thereby improving health behaviors among older adults and mitigating existing inequalities.

Building on this, the present study not only reveals the positive impact of adult children's education on the health behaviors of middle-aged and elderly parents, but also underscores the importance of upward spillover effects of human capital. The results suggest that parental investment in children's education generates broader returns beyond the younger generation. Such investment not only enhances children's knowledge accumulation and socioeconomic status, but also indirectly promotes healthier behaviors among parents, thereby increasing the overall value of human capital within the household. From a policy perspective, these findings provide a potential pathway for improving the health and well-being of aging populations: increasing investment in education may yield long-term intergenerational benefits by leveraging the spillover effects of human capital.

However, several limitations of this study suggest directions for future research. To further validate and deepen the conclusions, future work could employ alternative empirical approaches, such as quantile regression, to examine the robustness of the results across the distribution of health behaviors. In addition, exploring the role of children's age in shaping the health behaviors of older parents may provide richer insights into the dynamics of intergenerational influence. Addressing these issues would help refine our understanding of how education shapes health outcomes within families.

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Appendix I

Table A1 The OLS regression results for Eq.1

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Smoke	Drink	Doctor	Siesta	Cleanwater
ChildEduy	-0.0053***	-0.0010	0.0042***	0.0056***	0.0101***
	(0.0008)	(0.0007)	(0.0012)	(0.0011)	(0.0013)
age	0.0207***	0.0109**	0.0234***	-0.0235***	0.0020
	(0.0050)	(0.0043)	(0.0075)	(0.0063)	(0.0070)
agesquare	-0.0002***	-0.0001***	-0.0002***	0.0002***	-0.0000
	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
marriage	0.0134***	0.0032	-0.0042	-0.0022	0.0121***
	(0.0031)	(0.0025)	(0.0040)	(0.0039)	(0.0040)
gender	0.5688***	0.2975***	-0.0261***	0.0295***	-0.0177***
	(0.0064)	(0.0056)	(0.0086)	(0.0064)	(0.0037)
eduy	-0.0030***	-0.0021***	-0.0025**	0.0052***	0.0055***
	(0.0008)	(0.0007)	(0.0010)	(0.0009)	(0.0009)
ChildNumber	0.0089***	-0.0010	0.0009	-0.0072*	-0.0216***
	(0.0032)	(0.0026)	(0.0039)	(0.0038)	(0.0046)
ChildGender	0.0024	0.0073	-0.0260***	-0.0052	-0.0302***
	(0.0066)	(0.0058)	(0.0094)	(0.0083)	(0.0092)
ChildAge	0.0004	0.0022***	0.0028***	0.0032***	0.0030***
	(0.0008)	(0.0007)	(0.0010)	(0.0010)	(0.0011)
urban	-0.0017	-0.0055	-0.0337***	0.0019	0.2129***
	(0.0066)	(0.0057)	(0.0091)	(0.0082)	(0.0099)
log(perincome)	0.0017	0.0056***	-0.0011	0.0149***	0.0244***
	(0.0022)	(0.0019)	(0.0035)	(0.0029)	(0.0034)
Province FE	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family
Observations	36,485	36,484	13,060	36,480	(0.2062)

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The reported results in the table are coefficients corresponding to the explanatory variables. For example, the number -0.0053 corresponding to *ChildEduy* in (1) indicates that, with other variables remaining unchanged, for each additional year of education for children, the probability of their parents smoking decreases by an average of 0.0053.

Table A2 Table of Heterogeneous Regression Results for *Smoke*

	Parental						Offspring's Gender	
	Age		Gender		Urban-Rural			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Middle-Aged	Young Elderly	Mother	Father	Rural	Urban	Daughter	Son
VARIABLES	Smoke	Smoke	Smoke	Smoke	Smoke	Smoke	Smoke	Smoke
ChildEduy	-0.0015	-0.0118	-0.0037	-0.0290*	-0.0243***	-0.0187	-0.0345**	-0.0188*
	(0.0092)	(0.0100)	(0.0068)	(0.0150)	(0.0094)	(0.0189)	(0.0149)	(0.0098)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,055	13,430	19,282	17,203	20,140	16,345	10,317	26,168
P-value	0.210		0.030		0.420		0.110	

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The P-value for the coefficient difference was obtained through the seemingly unrelated regression model test.

Table A3 Table of Heterogeneous Regression Results for Drink

	Parental						Offspring's Gender	
	Age		Gender		Urban-Rural			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Middle-Aged	Young Elderly	Mother	Father	Rural	Urban	Daughter	Son
VARIABLES	Drink	Drink	Drink	Drink	Drink	Drink	Drink	Drink
ChildEduy	-0.0008	0.0039	-0.0017	0.0024	0.0062	-0.0145	-0.0208	0.0081
	(0.0083)	(0.0088)	(0.0055)	(0.0130)	(0.0079)	(0.0168)	(0.0130)	(0.0084)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,054	13,430	19,282	17,202	20,140	16,344	10,316	26,168
P-value	0.330		0.290		0.030		0.020	

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The P-value for the coefficient difference was obtained through the seemingly unrelated regression model test.

Table A4 Table of Heterogeneous Regression Results for Doctor

	Parental						Offspring's Gender	
	Age		Gender		Urban-Rural			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Middle-Aged	Young Elderly	Mother	Father	Rural	Urban	Daughter	Son
VARIABLES	Doctor	Doctor	Doctor	Doctor	Doctor	Doctor	Doctor	Doctor
ChildEduy	0.0356**	0.0949***	0.0812***	0.0746***	0.0605***	0.1492**	0.0798**	0.0754***
	(0.0155)	(0.0222)	(0.0230)	(0.0208)	(0.0150)	(0.0761)	(0.0378)	(0.0176)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,674	5,386	8,007	5,053	7,617	5,443	3,434	9,626
P-value	0.030		0.440		0.000		0.470	

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The P-value for the coefficient difference was obtained through the seemingly unrelated regression model test.

Table A5 Table of Heterogeneous Regression Results for Siesta

	Parental						Offspring's Gender	
	Age		Gender		Urban-Rural			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Middle-Aged	Young Elderly	Mother	Father	Rural	Urban	Daughter	Son
VARIABLES	Siesta	Siesta	Siesta	Siesta	Siesta	Siesta	Siesta	Siesta
ChildEduy	0.0075	0.0351***	0.0106	0.0202	0.0323***	-0.0132	0.0051	0.0211*
	(0.0141)	(0.0125)	(0.0145)	(0.0132)	(0.0121)	(0.0232)	(0.0188)	(0.0124)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,051	13,429	19,279	17,201	20,138	16,342	10,315	26,165
P-value	0.060		0.350		0.000		0.270	

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The P-value for the coefficient difference was obtained through the seemingly unrelated regression model test.

Table A6 Table of Heterogeneous Regression Results for Cleanwater

	Parental						Offspring's Gender	
	Age		Gender		Urban-Rural			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Middle-Aged	Young Elderly	Mother	Father	Rural	Urban	Daughter	Son
VARIABLES	Cleanwater	Cleanwater	Cleanwater	Cleanwater	Cleanwater	Cleanwater	Cleanwater	Cleanwater
ChildEduy	0.0518***	0.0728***	0.0662***	0.0612***	0.0972***	-0.0052	0.0331*	0.0701***
	(0.0170)	(0.0161)	(0.0159)	(0.0140)	(0.0189)	(0.0231)	(0.0197)	(0.0167)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,583	14,015	19,855	17,743	20,872	16,726	10,540	27,058
P-value	0.110		0.390		0.000		0.020	

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The P-value for the coefficient difference was obtained through the seemingly unrelated regression model test.

Table A7 Robustness Test - Regression Results Table for Logit Model

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Smoke	Drink	Doctor	Siesta	Cleanwater
ChildEduy	-0.0053***	-0.0012*	0.0043***	0.0056***	0.0095***
	(0.0008)	(0.0007)	(0.0012)	(0.0011)	(0.0012)
Control variables	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family
Observations	36,483	36,473	13,056	36,478	37,595

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the dependent variable takes a value of 1, it indicates that the behavior has been performed. The reported results in the table are average marginal effects. For example, the corresponding number of -0.0053 for ChildEduy in (1) indicates that, with other variables held constant, the probability of parents smoking decreases by 0.53% for each additional year of their children's education.

Table A8 Robustness Test - Regression Results Table of Adding Control Variables

Table A8a Smoke, Drink and Doctor

	Smoke		Drink		Doctor	
	(1)	(2)	(3)	(4)	(5)	(6)
	IV-2SLS	First-stage	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Smoke	ChildEduy	Drink	ChildEduy	Doctor	ChildEduy
ChildEduy	-0.0186**		0.0065		0.0621***	
	(0.0087)		(0.0075)		(0.0154)	
selfhealth	-0.0082***	0.0162	-0.0168***	-0.0161***	0.0505***	-0.0396
	(0.0019)	(0.0189)	(0.0017)	(0.0189)	(0.0037)	(0.0288)
Exposure		1.8509***		1.8523***		1.7455***
		(0.2032)		(0.2032)		(0.2624)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family	Family
Minimum eigenvalue	306.801		307.209		103.907	
Observations	36,481	36,481	36,480	36,480	13,059	13,059

Table A8b Siesta and Cleanwater

	Siesta		Cleanwater	
	(7)	(8)	(9)	(10)
	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Siesta	ChildEduy	Cleanwater	ChildEduy
ChildEduy	0.0117		0.0120***	
	(0.0110)		(0.0031)	
selfhealth	0.0157***	0.0162	-0.0066***	-0.0788***
	(0.0024)	(0.0189)	(0.0023)	(0.0199)
Exposure		1.8517***		4.0060***
		(0.2032)		(0.1215)
Control variables	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family
Minimum eigenvalue	306.993		5,276.21	
Observations	36,476	36,476	41,035	41,035

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the value of the dependent variable in column (1)(3)(5)(7)(9) is 1, it indicates that this behavior has been performed. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.

Table A9 Robustness Test - Regression Results Table of Adjusting the Sample Period

Table A9a Smoke, Drink and Doctor

	Smoke		Drink		Doctor	
	(1)	(2)	(3)	(4)	(5)	(6)
	IV-2SLS	First-stage	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Smoke	ChildEduy	Drink	ChildEduy	Doctor	ChildEduy
ChildEduy	-0.0195**		0.0053		0.0856***	
	(0.0094)		(0.0083)		(0.0227)	
Exposure		1.7871***		1.7878***		1.5279***
		(0.1980)		(0.1980)		(0.2724)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family	Family
Minimum eigenvalue	187.142		187.253		50.6374	
Observations	20,937	20,937	20,936	20,936	7,445	7,445

Table A9b Siesta and Cleanwater

	Siesta		Cleanwater	
	(7)	(8)	(9)	(10)
	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Siesta	ChildEduy	Cleanwater	ChildEduy
ChildEduy	0.0312**		0.0166***	
	(0.0128)		(0.0033)	
Exposure		1.7863***		3.9920***
		(0.1980)		(0.1196)
Control variables	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family
Minimum eigenvalue	186.932		3,260.09	
Observations	20,935	20,935	24,124	24,124

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the value of the dependent variable in column (1)(3)(5)(7)(9) is 1, it indicates that this behavior has been performed. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.

Table A10 Robustness Test - Regression Results Table of Adjusting the Upper Age Limit for Parents

Table A10a Smoke, Drink and Doctor

	Smoke		Drink		Doctor	
	(1)	(2)	(3)	(4)	(5)	(6)
	IV-2SLS	First-stage	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Smoke	ChildEduy	Drink	ChildEduy	Doctor	ChildEduy
ChildEduy	-0.0170**		0.0036		0.0688***	
	(0.0080)		(0.0069)		(0.0146)	
Exposure		1.9156***		1.9168***		1.7978***
		(0.1937)		(0.1938)		(0.2487)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family	Family
Minimum eigenvalue	364.923		365.308		121.607	
Observations	38,440	38,440	38,439	38,439	13,882	13,882

Table A10b Siesta and Cleanwater

	Siesta		Cleanwater	
	(7)	(8)	(9)	(10)
	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Siesta	ChildEduy	Cleanwater	ChildEduy
ChildEduy	0.0138		0.0098***	
	(0.0102)		(0.0026)	
Exposure		1.9161***		3.9920***
		(0.1938)		(0.1196)
Control variables	Yes	Yes	Yes	Yes
Province fFe	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family
Minimum eigenvalue	365.024		6,713.08	
Observations	38,435	38,435	43,846	43,846

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the value of the dependent variable in column (1)(3)(5)(7)(9) is 1, it indicates that this behavior has been performed. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.

Table A11 Robustness Test - Regression Results Table of Replacing the Core Explanatory Variable

Table A11a Smoke, Drink and Doctor

	Smoke		Drink		Doctor	
	(1)	(2)	(3)	(4)	(5)	(6)
	IV-2SLS	First-stage	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Smoke	ChildEduy	Drink	ChildEduy	Doctor	ChildEduy
ChildEduy	-0.0711***		-0.0041		0.2598***	
	(0.0273)		(0.0232)		(0.0589)	
Exposure		0.5804***		0.5809***		0.5064***
		(0.0655)		(0.0655)		(0.0836)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family	Family	Family
Minimum eigenvalue	254.421		254.766		76.0529	
Observations	38,002	38,002	38,001	38,001	13,663	13,663

Table A11b Siesta and Cleanwater

	Siesta		Cleanwater	
	(7)	(8)	(9)	(10)
	IV-2SLS	First-stage	IV-2SLS	First-stage
VARIABLES	Siesta	ChildEduy	Cleanwater	ChildEduy
ChildEduy	0.0501		0.0365***	
	(0.0343)		(0.0091)	
Exposure		0.5808***		1.3207***
		(0.0655)		(0.0389)
Control variables	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Cluster	Family	Family	Family	Family
Minimum eigenvalue	254.731		4,805.67	
Observations	37,997	37,997	43,352	43,352

Note: The robust standard errors of are reported in parentheses. Significance levels are indicated by ***, **, and * for the 1%, 5%, and 10% levels, respectively. When the value of the dependent variable in column (1)(3)(5)(7)(9) is 1, it indicates that this behavior has been performed. The Minimum eigenvalue statistic is used to test whether the model has a weak instrumental variable problem. If this statistic is greater than the 10% critical value of 16.38 for the 2SLS Size of nominal 5% Wald test, it indicates that the model does not have a weak instrumental variable problem.